

FEDERAL BOARD OF INTERMEDIATE & SECONDARY EDUCATION**Annual Examination 2026****Physics — Class 9 (Science Group)**

Syllabus: Unit 1 — Physical Quantities & Measurement | Unit 2 — Kinematics | Unit 3 — Dynamics-I | Unit 4 — Dynamics-II | Unit 5 — Pressure & Deformation in Solids | Unit 6 — Work & Energy (till 6.4) | Unit 9 — Nature of Science and Physics

Total Marks: 46**Time: 2 Hours****Paper: Exam-15 (Version 1)**

1. Attempt ALL questions. There is no choice or OR alternative anywhere in this paper.
2. Section A (30 marks): 30 MCQs. Fill the correct bubble. Suggested time: 35 minutes.
3. Show complete working for every calculation. Always write the correct unit with your final answer.
4. Use black or blue ink only. Pencil is allowed only for diagrams. Take $g = 9.8 \text{ N/kg}$ on Earth unless a different value is given in the question.

SECTION A — Multiple Choice Questions ($30 \times 1 = 30$ Marks)

#	Q.1 Choose the best possible option.	A	B	C	D	Answer
1	The number of significant figures in the measurement 0.02080 kg is:	2	3	4	5	A B C D
2	An archer shoots several arrows at a target. All the arrows land close together, forming a tight cluster, but far from the bullseye. This set of results is:	Accurate but not precise	Precise but not accurate	Both precise and accurate	Neither precise nor accurate	A B C D
3	Which instrument would you use to measure the WEIGHT of a body directly, rather than its mass?	Physical balance	Spring balance (force meter)	Measuring cylinder	Vernier caliper	A B C D
4	A small stone is lowered into a measuring cylinder containing water, and the water level rises from 42 mL to 65 mL. The volume of the stone is:	23 cm ³	42 cm ³	65 cm ³	107 cm ³	A B C D
5	A ball is dropped from rest and falls freely under gravity. If it reaches a velocity of 35 m/s, the time taken to reach this velocity (taking $g = 10 \text{ m/s}^2$) is:	0.35 s	7 s	3.5 s	350 s	A B C D
6	The distance-time graph of a car is a straight horizontal line parallel to the time axis. This means the car is:	At rest	Moving with constant speed	Accelerating uniformly	Decelerating uniformly	A B C D
7	A train's speed increases uniformly from 10 m/s to 40 m/s in 15 s, as recorded on a speed-time graph. The GRADIENT of this graph represents the train's:	Acceleration, equal to 2 m/s ²	Distance covered, equal to 375 m	Average speed, equal to 25 m/s	Displacement, equal to 600 m	A B C D
8	A car travels at a constant speed of 12 m/s for 25 s, as shown on a speed-time graph. The AREA under this graph represents the car's:	Average speed, equal to 12 m/s	Distance travelled, equal to 300 m	Acceleration, equal to 0.48 m/s ²	Change in velocity, equal to 0 m/s	A B C D
9	A trolley of mass 0.5 kg, moving at 20 m/s, is brought to rest in 0.05 s by a retarding force. The impulse delivered to the trolley is:	0.5 N·s	200 N·s	400 N·s	10 N·s	A B C D
10	A bullet of mass 0.01 kg, moving horizontally at 200 m/s, embeds itself	1 m/s	2 m/s	0.5 m/s	100 m/s	A B C D

	in a stationary wooden block of mass 1.99 kg. Using the law of conservation of momentum, the common velocity of the bullet-block system immediately after impact is:					
11	A resultant force of 18 N produces an acceleration of 2 m/s ² in a body. The mass of the body is:	4.5 kg	36 kg	9 kg	6 kg	A B C D
12	In Table 3.1 of the textbook, the interaction between a proton and an electron is given as an example under which fundamental force?	Strong nuclear force	Electromagnetic force	Weak nuclear force	Gravitational force	A B C D
13	A uniform seesaw is pivoted at its centre. A weight of 30 N is placed 2 m from the pivot on the left side. What weight must be placed 3 m from the pivot on the right side to balance the seesaw?	20 N	45 N	60 N	15 N	A B C D
14	The four engines of an aircraft all produce thrust in the same forward direction. The forces exerted by the engines are an example of:	Unlike parallel forces	Balanced forces	Action-reaction forces	Like parallel forces	A B C D
15	A skydiver falling through the air reaches a constant (terminal) velocity when:	Air resistance exceeds her weight	Her weight becomes zero	Her weight equals the air resistance (net force is zero)	Air resistance becomes zero	A B C D
16	Which of the following is a DISADVANTAGE of friction, according to the textbook?	Helps us walk (shoe-ground friction)	Causes wear and tear of moving parts	Helps a car's tyres grip the road	Holds nails and screws in wood	A B C D
17	A well-type manometer containing a liquid of density 1200 kg/m ³ shows a height difference of 15 cm between its two levels. Taking g = 10 m/s ² , the pressure difference indicated by the manometer is:	180 Pa	18,000 Pa	12,000 Pa	1800 Pa	A B C D
18	A mass of 4 kg is hung from a spring, causing it to extend by 2 cm. Taking g = 10 m/s ² , the spring constant of the spring is:	200 N/m	500 N/m	2000 N/m	800 N/m	A B C D
19	One pascal (Pa), the SI unit of pressure, is defined as a pressure of:	1 N/m ²	1 N/m ³	1 N·m	1 kg/m ²	A B C D
20	In a hydraulic press, the input piston has a cross-sectional area of 0.0015 m ² and the output piston has an area of 0.45 m ² . If a force of 20 N is applied to the input piston, the force delivered by the output piston is:	20 N	300 N	450 N	6000 N	A B C D
21	According to the textbook, the three common types of manometer are:	Mercury, Water, and Oil manometers	U-tube, Well Reservoir, and Inclined manometers	Simple, Compound, and Differential manometers	U-tube, Bourdon, and Digital manometers	A B C D
22	A ball of mass 4 kg is dropped from a height of 10 m. Taking g = 10 m/s ² and ignoring air resistance, the	40 J	200 J	400 J	800 J	A B C D

	kinetic energy of the ball just before it hits the ground is:					
23	A worker pulls a loaded sledge across a snowy field with a force of 90 N applied at an angle of 60° to the horizontal, moving it a distance of 3 m. Using $W = Fd \cos\theta$, the work done is:	135 J	270 J	45 J	233.8 J	A B C D
24	Which of the following is classified as a RENEWABLE energy resource?	Natural gas	Uranium	Petroleum	Geothermal heat	A B C D
25	An electric motor lifts a load of mass 200 kg through a height of 6 m in 10 s. Taking $g = 10 \text{ m/s}^2$, the power developed by the motor is:	120 W	200 W	1200 W	12,000 W	A B C D
26	Which of the following materials is used as fuel in nuclear power plants, according to the textbook?	Carbon-12	Uranium-235	Helium-4	Nitrogen-14	A B C D
27	According to the Physics textbook's Key Points, technology is best defined as:	A systematic body of knowledge about the universe	The process of studying and developing technology	A tentative explanation made before research	Tools/methods that help people accomplish tasks	A B C D
28	Which branch of physics deals with the study of the Earth's climate?	Climate Physics	Acoustics	Geophysics	Computational Physics	A B C D
29	Which of the following best defines a 'hypothesis' as used in the Physics textbook?	A pattern that summarises a regularity in nature	A well-supported explanation of a natural phenomenon	A tentative explanation made before research, later verified	An idea repeatedly tested and never disproven	A B C D
30	Physics is considered a branch of which broader category of science?	Biological Sciences	Social Sciences	Life Sciences	Physical Sciences	A B C D

SECTION B — Short Questions ($2 \times 4 = 8$ Marks)

Attempt ALL questions. There is no choice or OR alternative anywhere in this section.

Part	Question	Marks
2(i)	Differentiate between a scientific theory and a scientific law , giving two examples of each, as described in the textbook.	4
2(ii)	A rock sample has a mass of 55 kg. (a) Calculate its weight on Earth, taking $g = 9.8 \text{ N/kg}$. (b) Calculate its weight on the Moon, where the gravitational field strength is 1.62 N/kg . (c) State whether the mass of the rock changes when it is taken to the Moon, giving a reason.	4

SECTION C — Long Question ($1 \times 8 = 8$ Marks)

Attempt this question. There is no choice or OR alternative.

Part	Question	Marks
3	A satellite orbits the Earth in a circular path of radius 6800 km, completing one revolution in 5440 s. (a) Define orbital motion, and state the formula used to calculate the average orbital speed of a satellite. [2] (b) Calculate the satellite's orbital speed in m/s. [2] (c) Convert this speed into km/h. [1] (d) Name the force that provides the necessary centripetal force for the satellite's circular motion, and state its direction. [1] (e) If the satellite has a mass of 750 kg, calculate the magnitude of the centripetal force acting on it, using $F = mv^2/r$. [2]	8

Student's Signature: _____

Invigilator: _____

Date: _____

Chapters Covered: Unit 1 – Physical Quantities & Measurement, Unit 2 – Kinematics, Unit 3 – Dynamics-I, Unit 4 – Dynamics-II, Unit 5 – Pressure & Deformation in Solids, Unit 6 – Work & Energy (till 6.4), Unit 9 – Nature of Science and Physics

— End of Question Paper —